Probability:

At its core, **Maximum Likelihood Estimation (MLE)** is a method for finding the "best fit" parameters for a dataset. For a **Normal Distribution**, this means **finding the specific mean (μ) and variance (σ^2) that make your observed data most likely to have occurred**.

Here is a step-by-step breakdown of the algorithm and the logic behind it:

1. The Likelihood Function

Assume you have a set of independent data points ($x_1, x_2, ...x_n$). The probability of seeing all these points together is the product of their individual probabilities (the **Probability Density Function**).

For a Normal Distribution, the likelihood function L is:

$$L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$$

2. The Log-Likelihood (The "Trick")

Differentiating a long product is difficult. To simplify, we take the **natural logarithm** ($\ln$) of the function. Because the logarithm is a "monotonically increasing" function, the μ and σ that maximize the log-likelihood will also maximize the original likelihood.

Taking the log turns the products into **sums** and brings down the exponents:

$$\ln L(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \sum_{i=1}^{n}\frac{(x_i - \mu)^2}{2\sigma^2}$$

3. Maximizing the Function

To find the maximum, we use calculus: take the **partial derivative** with respect to μ and σ^2, set them to **zero**, and solve.

- **For the Mean (μ):** When you derive and solve, you find that the MLE for the mean is simply the **sample average**:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- **For the Variance (σ^2):** When you solve for σ^2, you find it is the **average squared distance** from the mean:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{\mu})^2$$